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MATHEMATICS TEST - CHAPTER 4: FACTORIZATION AND ALGEBRAIC MANIPULATION

Name:		Roll No.		Section:

Total Time: 2:30 Hour **Total Marks: 150**
Time Allowed: 150 Minutes

PAPER OBJECTIVE
SECTION-A (MARKS 150)

NOTE: Section-A is compulsory. All parts of this section are to be answered on these pages and handed over to the center superintendent. Deleting / overwriting is not allowed. Do not use lead pencil.

Q.1 Fill the relevant bubble for each part. All parts carry one mark.

Questions	A	B	C	D	A B C D
1. The process of expressing an expression as a product of its factors is called	Expansion	Simplification	Substitution	Factorization	○○○●
2. Factorization is essentially the opposite of	Addition	Subtraction	Division by zero	Multiplication	○○○●
3. ka + kb + kc factorizes as	k(a + b) + c	k(a + b + c)	a(k + b + c)	k + a + b + c	○●○○
4. ac + ad + bc + bd factorizes as	(a + c)(b + d)	(c + d)(a + b)	(a + d)(b + d)	abcd	○●○○
5. $a^2 \pm 2ab + b^2$ equals	$(a \pm b)^2$	$a^2 \pm b^2$	$(a \pm b)(a \mp b)$	$2(a \pm b)$	●○○○
6. $a^2 - b^2$ equals	$(a - b)^2$	$(a + b)^2$	$a(a - b)$	$(a + b)(a - b)$	○○○●
7. $a^2 + 2ab + b^2 - c^2$ factorizes as	$(a + b + c)(a + b - c)$	$(a + b - c)^2$	$(a - b)(a + c)$	$(a + b)^2 + c^2$	●○○○
8. $a^4 + a^2b^2 + b^4$ is factorized by adding and subtracting	$2a^2b^2$	a^2b^2	a^4	b^4	○●○○
9. $a^4 + a^2b^2 + b^4$ equals	$(a^2 + b^2 + ab)(a^2 + b^2 - ab)$	$(a^2 + b^2)^2$	$(a^2 + ab + b^2)^2$	$(a^2 - b^2)(a^2 + b^2)$	●○○○
10. $a^4 + 4b^4$ is factorized by adding and subtracting	$2a^2b^2$	a^4	$4a^2b^2$	$4b^4$	○○●○
11. $a^4 + 4b^4$ equals	$(a^2 + 2b^2)^2$	$(a^2 + 4b^2)(a^2 - 4b^2)$	$(a + 2b)^4$	$(a^2 + 2b^2 + 2ab)(a^2 + 2b^2 - 2ab)$	○○○●
12. To factorize $x^2 + px + q$, we look at the product of	Middle coefficients only	Extreme coefficients	Constants only	All coefficients	○●○○
13. If the middle term is negative and last term positive, both factors are	Negative	Positive	One of each sign	Zero	●○○○
14. If middle and last terms are both negative, the factors are	Both positive	One positive and one negative	Both negative	Both zero	○●○○

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Questions	A	B	C	D	A B C D
15. For $a^3 + 3a^2b + 3ab^2 + b^3$ the factorization is	$(a - b)^3$	$(a + b)^3$	$a^3 + b^3$	$(a + b)(a^2 + b^2)$	○●○○
16. For $a^3 - 3a^2b + 3ab^2 - b^3$ the factorization is	$(a + b)^3$	$a^3 - b^3$	$(a - b)(a^2 - b^2)$	$(a - b)^3$	○○○●
17. $a^3 + b^3$ equals	$(a + b)(a^2 + ab + b^2)$	$(a + b)(a^2 - ab + b^2)$	$(a + b)^3$	$(a - b)(a^2 + ab + b^2)$	○●○○
18. $a^3 - b^3$ equals	$(a - b)(a^2 - ab + b^2)$	$(a - b)^3$	$(a - b)(a^2 + ab + b^2)$	$(a + b)(a^2 - ab + b^2)$	○○●○
19. $x^6 - y^6$ is first written as	$(x^2)^3 + (y^2)^3$	$(x - y)^6$	$(x^4)^2 - (y^4)^2$	$(x^3)^2 - (y^3)^2$	○○○●
20. Factorize $x^2 + x - 72$	$(x + 9)(x - 8)$	$(x + 10)(x - 9)$	$(x + 9)(x + 8)$	$(x - 9)(x - 8)$	●○○○
21. Factorize $12x^2 - 41x + 35$	$(4x + 7)(3x - 5)$	$(3x - 7)(4x - 5)$	$(4x - 7)(3x + 5)$	$(4x - 7)(3x - 5)$	○○○●
22. $x^3 - 18x^2 + 108x - 216 =$	$(x - 6)^3$	$(x - 36)^3$	$(x - 6)(x^2 + 36)$	$(x + 6)^3$	●○○○
23. Factorize $4x^2 - 64$	$(2x + 8)(2x + 8)$	$(2x + 64)(2x - 64)$	$(2x - 8)(2x - 8)$	$(2x + 8)(2x - 8)$	○○○●
24. Factorize $15x^2 - 16x - 7$	$(5x - 7)(3x + 1)$	$(5x + 7)(3x + 1)$	$(5x + 7)(3x - 1)$	$(3x - 7)(5x + 1)$	●○○○
25. Factorize $x^2 - 5x - 24$	$(x + 3)(x - 8)$	$(x + 4)(x - 9)$	$(x + 3)(x + 8)$	$(x - 3)(x + 8)$	●○○○
26. Factorize $2x^2 + 13x - 7$	$(2x + 1)(x + 7)$	$(2x - 1)(x + 7)$	$(2x - 1)(x - 7)$	$(x - 1)(2x + 7)$	○●○○
27. Factorize $x^3 - 343$	$(x + 7)(x^2 + 7x + 49)$	$(x - 7)(x^2 + 7x + 49)$	$(x - 7)^3$	$(x - 7)(x^2 - 7x + 49)$	○●○○
28. Factorize $x^3 + 64$	$(x + 4)(x^2 + 4x + 16)$	$(x + 4)(x^2 - 8x + 16)$	$(x + 4)(x^2 - 4x + 16)$	$(x + 4)^3$	○○●○
29. Factorize $9x^2 - 12x + 4$	$(9x - 2)^2$	$(3x + 2)^2$	$(3x - 2)^2$	$(3x - 4)^2$	○○●○
30. Factorize $x^3 - 27$	$(x + 3)(x^2 + 3x + 9)$	$(x - 3)(x^2 + 3x + 9)$	$(x - 3)(x^2 + 6x + 9)$	$(x - 3)(x^2 - 3x + 9)$	○●○○
31. Factorize $x^2 - 25$	$(x - 5)(x - 5)$	$(x + 5)(x + 5)$	$(x - 5)(x + 5)$	$(x - 4)(x + 4)$	○○●○

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Questions	A	B	C	D	A B C D
32. Factorize $x^2 - 14x + 49$	$(x + 7)^2$	$(x - 7)^2$	$(x + 7)(x - 7)$	$(x - 49)^2$	○●○○
33. Factorize $8a^3 - 125b^3 - 2a + 5b$	$(2a - 5b)(4a^2 - 10ab + 25b^2 - 1)$	$(2a + 5b)(4a^2 + 10ab + 25b^2 - 1)$	$(2a - 5b)(4a^2 + 10ab + 25b^2 - 1)$	$(2a - 5b)(4a^2 + 10ab + 25b^2 + 1)$	○○●○
34. Factorize $4x^2 - x - 5$	$(4x + 5)(x - 1)$	$(x - 5)(4x + 1)$	$(4x - 5)(x + 1)$	$(4x - 5)(x - 1)$	○○●○
35. Factorize $27x^3 - 1$	$(3x - 1)(9x^2 + 3x + 1)$	$(3x + 1)(9x^2 - 3x + 1)$	$(3x - 1)^3$	$(3x - 1)(3x^2 + 3x + 1)$	●○○○
36. Factorize $5x^2 + 29x - 42$	$(5x + 6)(x - 7)$	$(5x + 6)(x + 7)$	$(5x - 6)(x + 7)$	$(x - 6)(5x + 7)$	○○●○
37. Factorize $x^2 + 2x - 63$	$(x - 6)(x + 8)$	$(x - 7)(x - 9)$	$(x + 7)(x + 9)$	$(x - 7)(x + 9)$	○○○●
38. Factorize $x^2 + 6x - 27$	$(x - 9)(x - 3)$	$(x - 9)(x + 3)$	$(x + 9)(x - 3)$	$(x + 10)(x - 4)$	○○●○
39. Factorize $8x^4 - 26x^2m^2 + 18m^4$	$2(2x^2 - m^2)(2x^2 - 9m^2)$	$2(x^2 + m^2)(4x^2 - 9m^2)$	$2(x^2 - m^2)(4x^2 - 9m^2)$	$2(x^2 - m^2)(4x^2 + 9m^2)$	○○●○
40. Factorize $9x^2 - 49$	$(3x + 7)(3x + 7)$	$(9x + 7)(9x - 7)$	$(3x - 7)(3x - 7)$	$(3x + 7)(3x - 7)$	○○○●
41. Factorize $x^2 + 13x + 36$	$(x + 9)(x + 4)$	$(x + 10)(x + 3)$	$(x - 9)(x - 4)$	$(x - 9)(x + 4)$	●○○○
42. Factorize $x^2 - 2x - 3$	$(x + 3)(x - 1)$	$(x - 3)(x + 1)$	$(x + 3)(x + 1)$	$(x - 3)(x - 1)$	○●○○
43. Factorize $5x^2 + 12x + 7$	$(5x + 7)(x - 1)$	$(5x - 7)(x - 1)$	$(x + 7)(5x + 1)$	$(5x + 7)(x + 1)$	○○○●
44. Factorize $x^2 + x - 2$	$(x + 1)(x + 2)$	$(x)(x + 1)$	$(x - 1)(x + 2)$	$(x + 1)(x - 2)$	○○●○
45. Factorize $5x^2 - 28x + 15$	$(5x + 3)(x - 5)$	$(5x - 3)(x + 5)$	$(x - 3)(5x - 5)$	$(5x - 3)(x - 5)$	○○○●
46. Factorize $y^2 - 7y + 12$	$(y + 3)(y + 4)$	$(y - 2)(y - 6)$	$(y - 3)(y - 4)$	$(y - 1)(y - 12)$	○○●○
47. Factorize $15x^2 - 11x + 2$	$(5x - 2)(3x + 1)$	$(5x + 2)(3x + 1)$	$(5x + 2)(3x - 1)$	$(5x - 2)(3x - 1)$	○○○●
48. Factorize $27a^3 + 1 + 27a^2 + 9a$	$(3a - 1)^3$	$(3a + 1)(9a^2 + 1)$	$(3a + 1)^3$	$(3a + 1)^2$	○○●○

Questions	A	B	C	D	A B C D
49. Factorize $x^2 + 4x - 5$	$(x - 5)(x + 1)$	$(x - 5)(x - 1)$	$(x + 5)(x - 1)$	$(x + 6)(x - 2)$	○○●○
50. Factorize $15x^2 - 17x - 4$	$(5x - 1)(3x - 4)$	$(3x + 1)(5x - 4)$	$(5x + 1)(3x + 4)$	$(5x + 1)(3x - 4)$	○○○●
51. Factorize $6x^2 - 11x + 4$	$(2x + 1)(3x + 4)$	$(2x - 1)(3x + 4)$	$(2x - 1)(3x - 4)$	$(2x + 1)(3x - 4)$	○○●○
52. Factorize $x^2 - 7x + 10$	$(x + 5)(x + 2)$	$(x - 5)(x + 2)$	$(x + 5)(x - 2)$	$(x - 5)(x - 2)$	○○○●
53. Factorize $5x^2 - 7x - 6$	$(x + 3)(5x - 2)$	$(5x - 3)(x - 2)$	$(5x + 3)(x + 2)$	$(5x + 3)(x - 2)$	○○○●
54. $x^3 + 12x^2 + 48x + 64 =$	$(x - 4)^3$	$(x + 16)^3$	$(x + 4)^3$	$(x + 4)(x^2 + 16)$	○○●○
55. Factorize $x^3 + 8$	$(x - 2)(x^2 + 2x + 4)$	$(x + 2)(x^2 - 4x + 4)$	$(x + 2)(x^2 + 2x + 4)$	$(x + 2)(x^2 - 2x + 4)$	○○○●
56. Factorize $x^2 - 36$	$(x - 6)(x - 6)$	$(x - 6)(x + 6)$	$(x + 6)(x + 6)$	$(x - 5)(x + 5)$	○●○○
57. Factorize $x^2 - 4x - 5$	$(x - 5)(x - 1)$	$(x - 4)(x)$	$(x - 5)(x + 1)$	$(x + 5)(x - 1)$	○○●○
58. Factorize $12x^2 + 29x + 14$	$(4x - 7)(3x + 2)$	$(4x + 7)(3x - 2)$	$(4x - 7)(3x - 2)$	$(4x + 7)(3x + 2)$	○○○●
59. Factorize $x^2 - 11x + 28$	$(x + 4)(x - 7)$	$(x - 4)(x - 7)$	$(x + 4)(x + 7)$	$(x - 3)(x - 8)$	○●○○
60. A common factor of two polynomials is a polynomial which	Divides them exactly	Is their sum	Is their product	Equals their difference	●○○○
61. To find the HCF of monomials, the HCF of numerical coefficients is found by	Prime factorization	Addition	Subtraction	Squaring	●○○○
62. In the HCF of monomials, a common variable is taken with its	Highest power	Lowest power	Sum of powers	Product of powers	○●○○
63. For compound expressions, HCF is the product of common factors with	Highest power	Least power	Zero power	Double power	○●○○
64. The key words for HCF are	Highest power and every factor	Highest power and common factor	Lowest power and every factor	Lowest power and common factor	○○○●
65. The key words for LCM are	Lowest power and common factor	Lowest power and every factor	Highest power and every factor	Highest power and common factor	○○●○

Questions	A	B	C	D	A B C D
66. HCF by division: the remainder becomes the	Next dividend only	Final answer at once	Next divisor	Quotient	○○●○
67. While finding HCF by division, we may multiply or divide by	A variable	Any polynomial	Zero	A non-zero constant	○○○●
68. HCF of P and Q is the same as HCF of mP and nQ where m, n are	Non-zero constants	Variables	Zero	Negative variables only	●○○○
69. The LCM of polynomials is the polynomial of	Lowest degree divisible by each	Highest degree dividing each	Degree zero	Highest degree only	●○○○
70. For two polynomials P and Q, HCF × LCM equals	P + Q	P − Q	P ÷ Q	P × Q	○○○●
71. LCM of P and Q can be found as	(P × Q) ÷ HCF	HCF ÷ (P × Q)	P + Q − HCF	P ÷ HCF + Q	●○○○
72. HCF of P and Q can be found as	LCM ÷ (P × Q)	(P × Q) ÷ LCM	P + Q − LCM	P ÷ LCM	○●○○
73. If HCF is x − 3, P = $x^2 - 5x + 6$ and LCM = $x^3 - 9x^2 + 26x - 24$, then Q = HCF × LCM ÷	LCM	P	HCF	P + LCM	○●○○
74. HCF of two polynomials $x^3 + 5x^2 + 6x$ and $x^3 + 9x^2 + 14x$ is	x + 2	$x^2 + 7x$	x(x + 3)	$x^2 + 2x$	○○○●
75. HCF is not affected by multiplying a polynomial with	A variable	x	A non-zero number	Any polynomial	○○●○
76. For LCM of monomials, the numerical coefficient part is the	HCF of coefficients	Sum of coefficients	LCM of coefficients	Product only	○○●○
77. LCM of monomials includes each variable with its	Lowest power	Sum of powers	Zero power	Highest power	○○○●
78. Find the HCF of 12b and $24a^2b^3$	288b	12b	$12a^2b^3$	$24a^2b^3$	○●○○
79. Find the HCF of $8ac^3$ and $6b^3c^2$	$3c^2$	$2c^2$	$2ab^3c^3$	$48c^2$	○●○○
80. Find the LCM of $10b^3c$ and $15a^3b^2c^3$	$5b^2c$	$35a^3b^3c^3$	$30a^3b^3c^3$	$5a^3b^3c^3$	○○●○
81. HCF of $a^3b - ab^3$ and $a^5b^2 - a^2b^5$ is	$a^2b(a - b)$	ab	ab(a − b)	ab(a + b)	○○●○
82. HCF of (x + 1)(x − 5) and (x + 1)(x − 4) is	(x − 1)	(x − 4)	(x − 5)	(x + 1)	○○○●

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Questions	A	B	C	D	A B C D
83. Find the HCF of $36ab^2c^4$ and $42a^4c^3$	$6a^4b^2c^4$	$6ac^3$	$7ac^3$	$1512ac^3$	○●○○
84. If HCF of P and Q is (x + 3) and LCM is (x + 3)(x + 5), then P × Q =	$(x + 3)(x + 5)$	$(x + 3)^2(x + 5)$	$(x + 3)^3(x + 5)$	$(x + 3)(x + 5)^2$	○●○○
85. HCF of $5(x^3 + 5x^2 + 6x)$ and $x^3 + 9x^2 + 14x$ is	$x^2 + 2x$	$5(x^2 + 2x)$	$x + 2$	$5x$	●○○○
86. HCF of $(x - 2)(x + 6)$ and $(x - 2)(x + 2)$ is	$(x - 2)$	$(x - 2)(x + 6)$	$(x + 6)$	$(x + 2)$	●○○○
87. If HCF of P and Q is 2 and LCM is 60, then P × Q =	62	120	30	58	○●○○
88. HCF of $(x + 1)(x - 1)$ and $(x + 1)(x + 8)$ is	$(x + 1)$	$(x + 8)$	$(x + 1)(x - 1)$	$(x + 1)(x - 1)(x + 8)$	●○○○
89. LCM of $(x - 8)(x + 2)$ and $(x - 8)(x - 5)$ is	$(x - 8)(x + 2)(x + 5)$	$(x + 2)(x - 5)$	$(x - 8)(x + 2)(x - 5)$	$(x - 8)(x + 2)$	○○●○
90. Find the LCM of $6ab^3c$ and $36a^2c^4$	$36a^2b^3c^4$	$6a^2b^3c^4$	$216a^2b^3c^4$	$36ac$	●○○○
91. If HCF of P and Q is (x + 2) and LCM is (x + 2)(x + 5), then P × Q =	$(x + 2)(x + 5)$	$(x + 2)^3(x + 5)$	$(x + 2)^2(x + 5)$	$(x + 2)(x + 5)^2$	○○●○
92. Find the HCF of $18a^3bc$ and $12b^2c^3$	$6bc$	$36a^3b^2c^3$	$216bc$	$36bc$	●○○○
93. Find the LCM of $18b^4c^3$ and $14a^4bc^3$	$252a^4b^4c^3$	$126a^4b^4c^3$	$128a^4b^4c^3$	$2a^4b^4c^3$	○●○○
94. Find the HCF of $12ab^2c$ and $24b^2c^2$	$12ab^2c^2$	$12b^2c$	$24b^2c$	$13b^2c$	○●○○
95. HCF of $x^2 - 64$ and $x^2 + 16x + 64$ is	$(x + 8)$	1	$x^2 - 64$	$(x - 8)$	●○○○
96. Find the LCM of $20a^3bc^4$ and $14a^4bc$	$2a^4bc^4$	$280a^4bc^4$	$140a^3bc$	$140a^4bc^4$	○○○●
97. HCF of $(x + 6)(x + 1)$ and $(x + 6)(x - 8)$ is	$(x + 6)(x + 1)$	$(x - 6)$	$(x + 6)$	$(x + 1)$	○○●○
98. If HCF = 3, LCM = 60 and P = 12, then Q =	15	5	48	2160	●○○○
99. LCM of $(x - 6)(x + 8)$ and $(x - 6)(x + 4)$ is	$(x - 6)(x + 8)(x + 4)$	$(x - 6)(x + 8)(x - 4)$	$(x + 8)(x + 4)$	$(x - 6)(x + 8)$	●○○○

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Questions	A	B	C	D	A B C D
100. HCF of $x^3 + x^2 - 5x + 3$ and $x^2 + 3x$ is	$x - 1$	$x + 3$	x	$x^2 + 3x$	○●○○
101. HCF of $(x + 2)(x - 1)$ and $(x + 2)(x - 6)$ is	$(x + 2)(x - 1)(x - 6)$	$(x - 1)$	$(x + 2)(x - 1)$	$(x + 2)$	○○○●
102. HCF of $x^2 - 36$ and $x^2 + 12x + 36$ is	$(x - 6)$	1	$x^2 - 36$	$(x + 6)$	○○○●
103. HCF of $(x - 2)(x - 1)$ and $(x - 2)(x + 3)$ is	$(x - 2)$	$(x - 2)(x - 1)$	$(x - 1)$	$(x - 2)(x - 1)(x + 3)$	●○○○
104. For P = 8 and Q = 20, HCF × LCM =	160	112	28	40	●○○○
105. HCF of $(x - 4)(x - 2)$ and $(x - 4)(x + 8)$ is	$(x + 8)$	$(x - 2)$	$(x - 4)$	$(x + 4)$	○○●○
106. HCF of $(x - 6)(x - 7)$ and $(x - 6)(x + 4)$ is	$(x - 6)$	$(x - 6)(x - 7)(x + 4)$	$(x - 6)(x - 7)$	$(x - 7)$	●○○○
107. Find the LCM of $10ab^2c$ and $18a^3b^4c^4$	$90ab^2c$	$2a^3b^4c^4$	$90a^3b^4c^4$	$92a^3b^4c^4$	○○●○
108. HCF of $x^2 - y^2$ and $(x - y)^3$ is	$x + y$	$x - y$	$(x - y)^2$	$x^2 - y^2$	○●○○
109. Find the LCM of $20a^2b^3c^3$ and $24a^4b^4c^3$	$120a^4b^4c^3$	$124a^4b^4c^3$	$4a^2b^3c^3$	$480a^4b^4c^3$	●○○○
110. If HCF of P and Q is $(x + 2)$ and LCM is $(x + 2)(x + 7)$, then P × Q =	$(x + 2)(x + 7)$	$(x + 2)^2(x + 7)$	$(x + 2)^3(x + 7)$	$(x + 2)(x + 7)^2$	○●○○
111. HCF of $(x + 3)(x + 5)$ and $(x + 3)(x - 4)$ is	$(x + 3)$	$(x + 3)(x + 5)$	$(x + 3)(x + 5)(x - 4)$	$(x - 4)$	●○○○
112. HCF of $(x - 4)(x + 4)$ and $(x - 4)(x - 3)$ is	$(x - 4)$	$(x - 4)(x + 4)$	$(x - 3)$	$(x - 4)(x + 4)(x - 3)$	●○○○
113. Expressions like $(x^2 + 2x + 1)/(2x^2 - 5x - 3)$ are called	Polynomials	Algebraic fractions	Whole numbers	Exponents	○●○○
114. (R/T) × (S/U) equals	RU/TS	RS/TU	R+S/T+U	RT/SU	○●○○
115. (S/U) ÷ (T/Q) equals	ST/UQ	SU/TQ	S/U − T/Q	SQ/UT	○○○●
116. To divide one fraction by another we	Invert the dividend and multiply	Subtract them	Invert the divisor and multiply	Add their reciprocals	○○●○

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Questions	A	B	C	D	A B C D
117. In multiplication of fractions, the product of cancelled common factors is the	LCM of numerator and denominator	Product of the fractions	Zero	HCF of numerator and denominator	○○○●
118. The key to simplifying an algebraic fraction is the ability to	Add polynomials	Draw graphs	Take square roots only	Factorize polynomials	○○○●
119. To add or subtract unlike fractions, we use the	HCF of denominators	LCM of denominators	Product of numerators	Sum of denominators	○●○○
120. Fractions with the same denominators are called	Like fractions	Unlike fractions	Improper fractions	Complex fractions	●○○○
121. For combined operations, the order is	ADMS	SMAD	DMAS	MSAD	○○●○
122. In combined operations, division is performed	After addition	Before addition	After subtraction	Last always	○●○○
123. 3 − x is equal to	x − 3	x + 3	−(x − 3)	−(x + 3)	○○●○
124. The reciprocal of an algebraic fraction a/b (a, b ≠ 0) is	−a/b	a/b	b/a	−b/a	○○●○
125. The product of an algebraic fraction and its reciprocal is	0	−1	The fraction	1	○○○●
126. $(x^2 - 64)/(x - 8) \div (x + 8) =$	1	x	0	x + 8	●○○○
127. $(x^2 - 25)/(x - 5) \div (x + 5) =$	x	1	0	x + 5	○●○○
128. Simplify 1/(x + 3) + 1/(x + 4)	2/(2x + 7)	(x + 7)/((x + 3)(x + 4))	1/((x + 3)(x + 4))	(2x + 7)/((x + 3)(x + 4))	○○○●
129. Simplify $(x + 4)/(x^2 - 16) \times (x - 4)$	x + 4	x − 4	0	1	○○○●
130. Simplify $(x^2 - 5x + 6)/(x - 2)$	x − 2	x − 6	x − 3	3	○○●○
131. Simplify $(x^2 - 7x + 12)/(x - 3)$	4	x − 4	x − 3	x − 12	○●○○
132. $(x^2 - 4)/(x - 2) \div (x + 2) =$	x	1	0	x + 2	○●○○
133. Simplify $(x^2 + 8x + 15)/(x + 3)$	x + 5	x + 3	x + 15	x − 5	●○○○

Questions	A	B	C	D	A B C D
134. The product of $x^3 + 7x - 8$ and its reciprocal is	0	1	$x^3 + 7x - 8$	-1	○●○○
135. Simplify $(x^2 + 8x + 7)/(x + 1)$	$x + 7$	$x + 1$	$x - 7$	7	●○○○
136. Simplify $(x^2 + 9x + 20)/(x + 4)$	$x + 5$	$x - 5$	$x + 4$	$x + 20$	●○○○
137. The square root of an algebraic expression P is Q if	$Q = P^2$	$Q^2 = P$	$Q + P = 0$	$Q = 2P$	○●○○
138. If Q is a square root of P, then another square root of P is	2Q	$-Q$	Q^2	1/Q	○●○○
139. The square root of an algebraic expression consists of	Two additive inverse expressions	One expression only	Two equal expressions	Three expressions	●○○○
140. $\sqrt{(4a^2 + 12ab + 9b^2)}$ equals	$\pm(2a + 3b)$	$\pm(2a - 3b)$	$(2a + 3b)^2$	$\pm(4a + 9b)$	●○○○
141. Square root of an expression can be found by	Addition method only	Multiplication only	Factorization or division method	Graph method only	○○●○
142. In the factorization method for square roots, the expression is first written as a	Cube	Difference of cubes	Product of three terms	Complete square	○○○●
143. $\sqrt{(x^2 + 2xy + y^2)}$ =	$\pm(x - y)$	$x^2 + y^2$	$x + y$ only	$\pm(x + y)$	○○○●
144. $\sqrt{(x^4)}$ =	$\pm x$	$\pm x^4$	$\pm x^2$	x^8	○○●○
145. An expression is a perfect square if its square root is	A polynomial	A fraction with zero	Zero only	Irrational always	●○○○
146. Square root of $16x^2 + 40x + 25$ is	$\pm(4x - 5)$	$(4x + 5)^2$	$4x + 5$	$\pm(4x + 5)$	○○○●
147. Square root of $x^2 - 2 + 1/x^2$ is	$\pm(x + 1/x)$	$\pm(x^2 - 1/x^2)$	$(x - 1/x)^2$	$\pm(x - 1/x)$	○○○●
148. Square root of $16a^2$ is	$4a$	$\pm 4a^2$	$\pm 4a$	± 16	○○●○
149. $\sqrt{(x^2 - 16x + 64)}$ =	$\pm(x + 8)$	$\pm(x^2 - 8)$	$\pm(x - 8)$	$(x - 8)^2$	○○●○
150. Square root of $16x^4$ is	$\pm 4x^2$	$\pm 16x^2$	$\pm 4x$	$16x^2$	●○○○

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